

MCS-012

1. Explain the addressing modes with an example each.
2. Explain the wilkes control unit with the help of a diagram.
3. Explain half-adder and full-adder circuit with logic diagram and truth table.
4. Explain the role of parity bed in error detection with the help of an example.
5. Explain the functioning of Master slave flip-flop with the help of a diagram.
6. What is instruction cycle? How are different kinds of instruction interpreted?
7. Explain the role of any five registers used in a basic Computer.
8. What is cache memory? Why is it needed? Explain the direct mapping cache organisations with the help of a diagram.
9. What are differences between .COM and .EXE program
10. Explain the term microor operations. How is a micro operations different from instruction?
11. List any five characteristics of RISC machine.
12. What is multiplexer? Draw the truth table and logic diagram of 2 X 1 multiplexer and explain its working.
13. What is DMA? Why is it useful? Draw the block diagram of a DMA interface.
14. What is a RAID? Explain all levels of RAID.
15. Explain the concept of instruction pipelining with the help of a diagram.
16. Calculate the physical address for the following register value/offset in 8086 microprocessor.
17. Simplify the boolean function in SOP from using K-Map also draw logic diagram of the simplified function using AND, OR, NOT gates.